What's the Route: Automatic Bouldering Route Detection

Myron Ladyjenko

University of Guelph, 50 Stone Rd. East, Guelph, ON N1G 2W1, Canada

Abstract

Bouldering route setting is a creative and skill-intensive process that involves selecting and positioning climbing holds on a wall to design engaging routes of varying difficulty. This project explores how machine learning and computer vision can assist in identifying and generating climbing routes automatically. Building upon previous work, which used YOLOv8 and Mask R-CNN models for object detection, with K-Means and DBSCAN algorithms for route identification, this project improves the quality of route detection by supporting simple data augmentation and preprocessing techniques and focusing on the YOLOv11 model. The project explores different clustering algorithms for colour and route detection and compares the results. Further, it investigates approaches toward route generation using Graph Neural Networks (GNNs) and Transformer decoders. By leveraging machine learning models and automation, this project aims to create a robust pipeline for route detection and further analyze how the automation can assist human route setters in generating creative, feasible, and diverse climbing routes.

Keywords: Machine Learning, Bouldering, Computer Vision, Object Detection, Route Generation, Clustering

1. Introduction

1.1. Background

Bouldering is a form of indoor rock climbing where climbers ascend short walls (approximately 4–5 meters high) without ropes or harnesses. Each wall features a combination of climbing holds, which are small, shaped grips and larger volumes that define possible climbing paths. A *route* (or "problem") consists of a specific set of holds, typically marked by the same colour. Each route has 'start' and 'finish' holds (marked by the tape), which define the starting and final position for the climbers. Climbers follow these colored sequences of holds to complete routes of increasing difficulty.

1.2. Motivation

Traditionally, route setting relies heavily on the intuition and experience of professional setters. However, as climbing continues to grow rapidly in popularity and more gyms open worldwide, the demand for creative and balanced routes has increased. Designing such routes remains time-consuming, subjective, and sometimes inconsistent. The following factors heavily influence the motivation for this project:

Growing Demand. As the popularity of indoor climbing continues to rise worldwide, gyms must reset routes frequently to keep climbers engaged. This increases the pressure on route setters to create a high volume of high-quality problems on a regular basis. The growing demand highlights the need for automated tools that can help with streamlining parts of the route setting workflow.

Route Setting is Slow. Designing balanced and creative routes requires significant expertise, physical effort, and time. Each route may take several hours to conceptualize, set, and test. This process becomes even more time-consuming as climbers want creativity and reasonably frequent turnover. Automating parts of this workflow can reduce setters' workload.

Unused Data from Existing Routes. Climbing gyms set dozens of climbs weekly. After a few weeks the climbs get reset and 'forgotten'. The data for the previous routes is rarely analyzed for insights or used to support future route design. By leveraging large annotated datasets and machine learning techniques, it is possible to extract route structures, movement patterns, and design principles that can help with data-driven route creation.

These factors motivate the development of a computer vision-based system to assist in classifying, analyzing, and generating climbing routes automatically, supporting route setters in producing creative and consistent bouldering problems.

1.3. Problem Statement

Climbing gyms rely on skilled route setters to design engaging bouldering problems for climbers. However, as the sport continues to grow in popularity, creating diverse and balanced routes has become increasingly demanding. By leveraging thousands of labelled images of holds and existing climbing routes, this project aims to develop a system that can automatically identify and generate climbing routes from wall images. The project aims to achieve the following objectives:

- Detect and classify holds and volumes from wall images using object detection models.
- Cluster holds by colour and spatial proximity to identify potential routes.

- Generate new route suggestions using relational graph representations.

The goal of the system is to assist human route setters by providing creative inspiration, improving consistency, and streamlining the route-setting process.

1.4. Literature Review

In the past, there have been many studies about climbing hold detection and route classification, as well as generation. For the current project, the following literature was covered:

The paper 'Indoor Rock Climbing Wall Route Displayer' (1) gives insight into climbers' positions on the wall, and because of that, it allows route setters to enhance the experience of climbers. Part of the paper also addresses the problem of detecting holds from an image and creating new routes for someone to climb, which is the primary focus.

While considering different approaches to creating a model for hold detection, different models such as YoLo, Mask R-CNN, DeepLab, RESNET and SAM2 were considered. A resource was found: Bouldering Route Classification, which contains implementation for a Mask R-CNN model for classifying routes, which helped with a starting point for working with this model and gave some suggestions that were tried in our experiments.

In earlier work (Route Classification in Bouldering Gyms), YOLOv11 and Mask R-CNN were implemented for object detection, achieving good performance but facing issues such as false positives from wall tape and wall bolts. Clustering algorithms such as K-Means and DBSCAN were used to group holds by colour, though these methods faced challenges due to chalk-covered holds and poor lighting. The current work builds on previous results to build a model that is able to generate climbing routes. For my background knowledge, previous work and further understanding, the following papers were read:

The paper 'Graph Neural Networks in Classifying Rock Climbing Difficulties' (2) introduces GNN to predict the difficulty levels of climbing routes on the MoonBoard. By modelling relationships between holds and problems as a graph, the authors demonstrate that such a model outperforms traditional models, achieving strong classification accuracy.

Next, the paper 'Recurrent Neural Network for Moonboard Climbing Route Classification and Generation' (3) introduces a new method 'BetaMove', which involves preprocessing sequences of moves required to complete a climb and using that to generate new climbs and classify the difficulty of the route.

To improve image quality and achieve better hold detection and route classification results, adaptive denoising methods such as hybrid median–based filters to reduce excessive chalk and lighting inconsistencies will be explored, as suggested in the paper: 'Denoising algorithm using adaptive and modified decision-based filters for enhanced image quality' (4).

Another two papers that I read are the following: 'Comparing YOLOv8 and Mask R-CNN for instance segmentation in complex orchard environments' (5) and 'YOLO Object Detection Algorithms' (6). These papers/articles were analyzed for the purposes of having a deeper understanding of the differences between the YOLO and Mask R-CNN models: advantages and disadvantages. This gave an understanding of which one to use when and how.

2. Materials and Methods

2.1. Dataset Description and Preprocessing

The project uses a composite dataset (with the images of bouldering walls and holds) drawn from several public sources:

1. Dataset I
2. Dataset II

Each dataset contains three splits: train, test, and validation sets for training, testing, and evaluation of the machine learning model (i.e. YOLOv11). Also, a small custom dataset was collected from the local climbing gym, the Guelph Grotto, to assess the model. From the public datasets, the combined one was created with a total number of images being 2365. The dataset combined different images and angles of the routes to incorporate different shading, lighting and chalk marks. To create the combined dataset, approximately 600 images have been randomly selected from the first dataset and ensured that all of the holds that are on the images are not 'volumes' since the dataset only has 1 label - 'holds'. This ensured that there were no misclassified holds in the combined dataset, as the combined dataset has two labels: 'holds' and 'volumes'. Further, for the dataset preprocessing step, the details about the data were standardized to follow the YOLO format in order to be used by the YOLO model. For the non-annotated images, the Roboflow framework allowed for uploading custom images and using the annotation tool to draw bounding boxes and label the holds. Once all of the images are annotated, the platform allows for the extraction of the data in the appropriate format based on the model with rotation and resize augmentations. The format used is YOLOv11 TXT for the YOLO model.

The YOLOv11 TXT format has images and corresponding TXT files that state which class the hold belongs to and the dimensions of the bounding box. Each annotation file contains one line per hold in the format **class_id x_center y_center width height**, where all values are normalized relative to the image dimensions. Using such metadata files, we don't need one large JSON file, and we can have information such as image ID and bounding box information. Furthermore, such structured input (from the dataset) will be used to construct graph representations of climbing walls, where each detected hold becomes a node and spatial relationships between holds form the edges used by the Graph Neural Network (GNN) for route generation and analysis. In the dataset, some of the annotation files contained segmentation results for the hold instead of the bounding box, which could significantly impact the training process and robustness of the model. In order to eliminate this problem, there was a preprocessing step introduced for all of the data in the dataset to convert any segmentation mask, if present, into a bounding box.

As a final step, the combined dataset was split into the three sets: training, testing and validation. This was accomplished by doing random sampling without replacement. Examples of the dataset images:

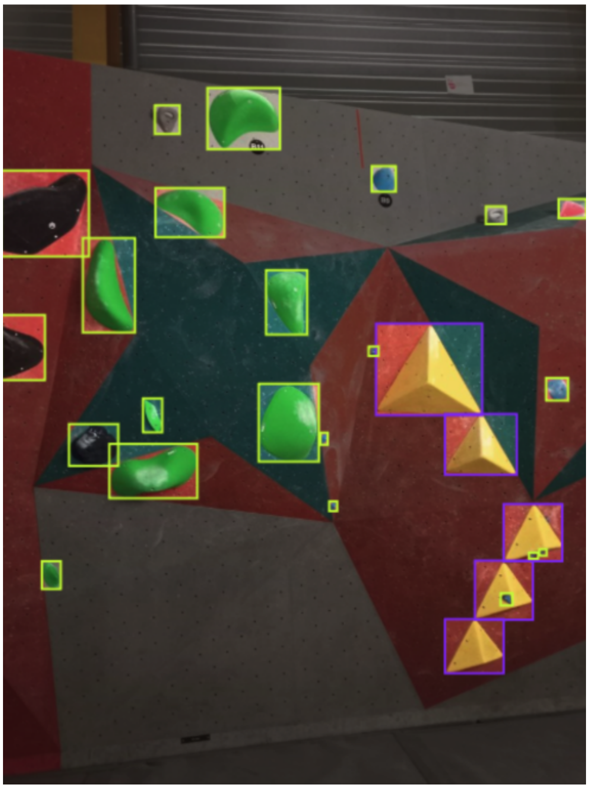


Figure 1: Example of the annotated image from the combined dataset.

Figure 2: Example of the annotated image from the combined dataset.

2.2. *System and Model Architecture*

Below, I will present the system architecture, split up into two major parts. The two parts below are the inference steps, which include the outline on the steps I will take to train the models. As a summary, the following languages, technologies and algorithms were applied for this project: **Python, PyTorch, K-Means, GMM, Mean-Shift, HUE HSV, DBSCAN, HDB-SCAN, Gaussian Blur, CLAHE**.

2.2.1. *Colour Extraction Algorithms*

- **K-Means:** A centroid-based clustering algorithm that partitions hold-pixel colours into a fixed number of clusters within the bounding box. Sensitive to lighting and chalk marks on the holds.

- **Mean-Shift:** A non-parametric algorithm that finds colour modes by sliding a window toward dense regions in the colour space. Handles shading, making it more robust for walls with variable lighting.

- **Gaussian Mixture Model (GMM):** A probabilistic clustering approach where each colour cluster is modelled as a Gaussian distribution. Provides soft assignments and handles overlapping colour regions.

- **Hue-Based HSV Peak Detection:** A histogram-based method that finds the dominant hue value within a hold's pixels. This technique is lighting-invariant but doesn't perform as well when saturation varies due to chalk.

2.2.2. *Route Clustering Algorithms*

- **DBSCAN:** A clustering algorithm that groups spatially close holds into clusters and labels isolated holds as noise (based on density). Effective for compact routes, but sensitive to tuning the parameters (distance to neighbours and minimum number of neighbours).

- **HDBSCAN:** An extension of DBSCAN that handles varying densities and automatically determines the number of clusters. It should be more robust for climbing walls where holds of the same route may be unevenly spaced.

- **Agglomerative Clustering:** A hierarchical method that iteratively merges the closest 'hold' clusters based on spatial distance. Useful when route holds must be grouped using a distance threshold rather than density.

2.3. *Part 1: Route Detection from Existing Wall Image*

This component takes a climbing wall image and automatically detects valid routes using the object detection model and clustering algorithms.

1. **Input:** User selects a part of the climbing wall image that contains different routes.
2. **Image Preprocessing:** Apply CLAHE contrast normalization and denoising to mitigate chalk on the holds and uneven lighting in the climbing gyms.

3. **Hold Detection:** Fine-tune a YOLO on the data to detect individual holds and use TTA (Test-Time Augmentation). Note that one option here is to try using YOLO segmentation model for higher precision.
4. **Colour Extraction** Perform colour-based clustering using a Gaussian Mixture Model (GMM), K-Means, Mean-Shift and Hue HSV to detect the colour of the hold. Once the colour for each hold is identified, group similar colours (i.e. close shades of red) into defined colour ranges. This is done so that each hold can be assigned to one of the ranges and thus will force the same colour holds, but due to lighting/chalk, different colour hex values, to belong to the same route, which will help with route detection.
5. **Route Detection** Use DBSCAN, Agglomerative clustering or HDBSCAN spatial clustering to identify candidate routes. The project will compare three different colour clustering and hold clustering algorithms to determine the best usage of them in different situations.
6. **Reachability Graph:** Build a graph where nodes are holds, and edges are human-feasible transitions based on distance and gravity-aware direction (i.e. hold is higher/lower).
7. **Sequence Extraction:** Infer a plausible climbing order via a shortest-path (or use some heuristic to select the next feasible hold).

For the above inference to work, the following algorithms will be implemented, and the models will be trained:

The climbing-wall images will be annotated with bounding boxes in YOLO format and split into training, validation, and test datasets. A YOLO model will then be trained and fine-tuned on this data for hold segmentation. After detection, the LAB colour features will be extracted from each predicted mask and fit a Gaussian Mixture Model (GMM) to cluster holds by colour (similarly, other algorithms will be implemented to extract the colour of the hold within the bounding box). Subsequently, the HDBSCAN algorithm will be applied to group spatially coherent holds within each colour cluster, producing candidate routes (the same will be done for the DBSCAN and Agglomerative clustering). Evaluation metrics (i.e. path validity score, human intervention) will be used to assess route quality and clustering accuracy.

2.4. Part 2: Route Generation

This component generates a new route (climb) by placing user-provided unpositioned holds onto an empty wall. Note that the hold types, their size and the wall angle are provided.

1. **Hold Input:** The user specifies a list of holds (i.e., jugs, crimps, volumes) and the wall angle (e.g., 10 degrees overhang).
2. **Graph Initialization:** Initialize an empty wall graph and embed hold properties (type, size and grip orientation).
3. **GNN Encoding:** A Graph Neural Network (GNN) learns spatial feasibility and human reach constraints over potential hold placements.
4. **Transformer Generation:** A Transformer decoder places holds one-by-one, generating both their positions and climb order.
5. **Feasibility Validation:** Enforce physical constraints (human reach, vertical progression) with a refinement from the route setters.

Similar to the above, for step 2, the following algorithms will be implemented, and the models will be trained:

Use a dataset that contains professionally set routes containing the wall angle, hold attributes (type, size and grip), holds placements, and the recommended climbed sequence of the holds. The dimensions in this dataset will be normalized and used in a GNN encoder that produces node/graph embeddings. Next, a Transformer decoder is trained with teacher forcing to predict the next hold placement, based on the already placed holds and the wall angle. The model will be evaluated on the accuracy of the sequence, the constraint-satisfaction rate and human ratings from route setters.

2.5. Reasoning for the approach

The reason for the approach that is outlined above lies in the purpose of the project. In the future, this system could be integrated into an application used by the climbing gyms. In the application, the system will be used to do inference in real-time on the images submitted from route setters (to detect routes on the wall), as well as suggest new climbs based on the wall angle and holds set during the route setting process. Due to that, it has been decided to use the YOLO model, since it is a single-stage detector with sufficient accuracy and a higher processing speed (requires the system fewer resources). Also, if the YOLO segmentation model is used, it will allow for pixel-level hold boundaries, which gives more accurate hold centers, sizes, and masks for the future graph construction. To have a better detection accuracy, I will try to apply different image processing techniques to enhance image quality and reduce noise. Furthermore, this approach gives the flexibility to explore different unsupervised clustering algorithms for colour and route detection, from which we can learn the limitations of certain algorithms and their strong sides in the domain of our problem.

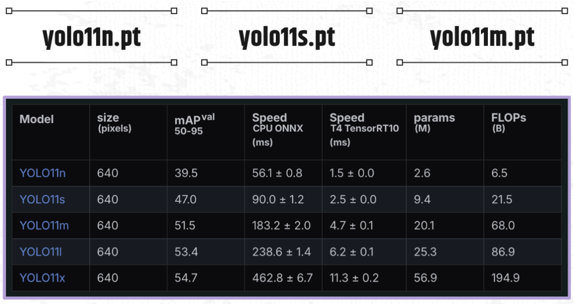
The reason behind using the GNN is that climbing routes can be modelled with a graph with certain constraints (i.e. distance between holds, direction, etc.). Such graphs are needed to account for the climber's possible moves and build realistic routes, and GNN can be used to learn such spatial relationships. Lastly, for generating new routes, a transformer-based decoder will be used since it can sequentially place holds while capturing spatial dependencies between holds and possible movement patterns better than heuristic-based algorithms.

3. Results

Below, I will provide results that were accomplished by this project, and further down, in the discussion section, the explanation and meaning for each of the results will be provided.

3.1. YOLOv11 Model Comparison

The images below demonstrate the efficacy of training the models for a longer amount of time and the difference in accuracy and training time between YOLO model sizes.



Model	size (pixels)	mAP^val 50-95	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	params (M)	FLOPs (B)
YOLO11n	640	39.5	56.1 ± 0.8	1.5 ± 0.0	2.6	6.5
YOLO11s	640	47.0	90.0 ± 1.2	2.5 ± 0.0	9.4	21.5
YOLO11m	640	51.5	183.2 ± 2.0	4.7 ± 0.1	20.1	68.0
YOLO11l	640	53.4	238.6 ± 1.4	6.2 ± 0.1	25.3	86.9
YOLO11x	640	54.7	462.8 ± 6.7	11.3 ± 0.2	56.9	194.9

Figure 3: Comparison of the different YOLOv11 models and their performance

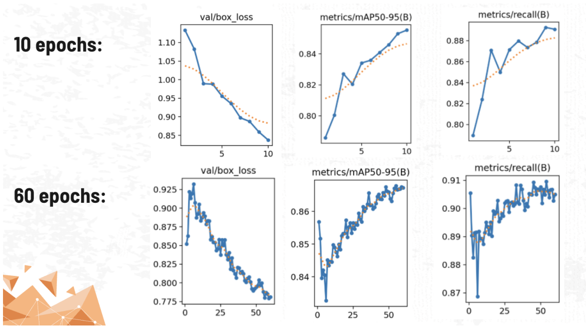


Figure 4: Comparison between training yolov11n, 10 and 60 epochs

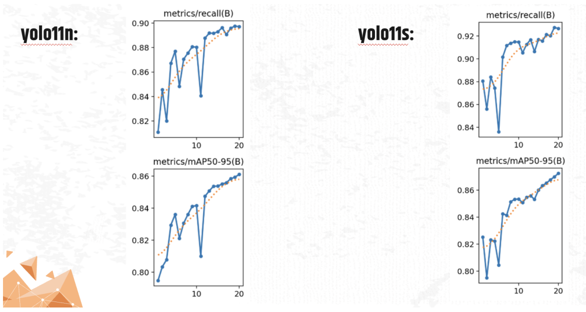


Figure 5: Comparison between nano and small YOLOv11 model

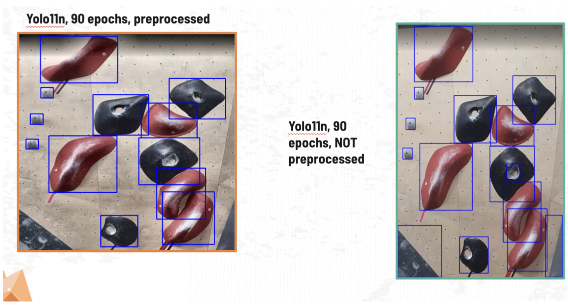


Figure 6: Comparison between detections on the preprocessed image vs raw

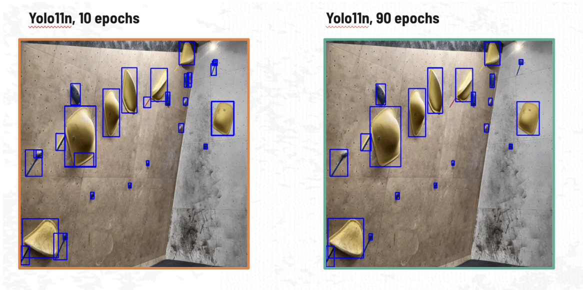


Figure 7: Hold detection with YOLOv11 nano: 10 and 90 epochs

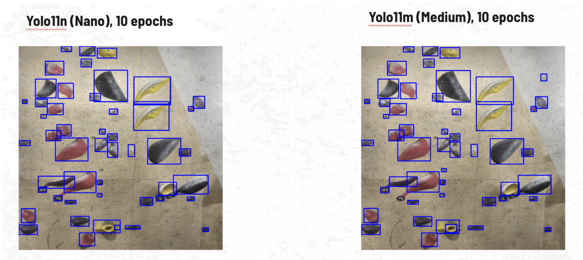


Figure 8: Hold detection with YOLOv11 nano vs medium (10 epochs)

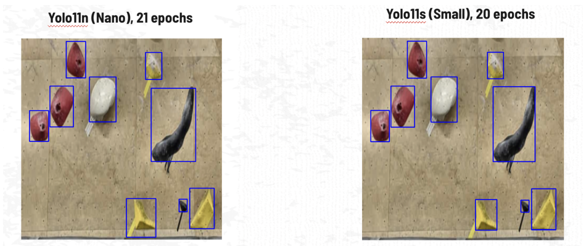


Figure 9: Hold detection with YOLOv11 nano vs medium (20 epochs)

3.2. Preprocessing

The preprocessing techniques (CLAHE contrast normalization and Gaussian blurring) improved image quality to eliminate poor lighting and chalk marks. After preprocessing, YOLOv11 achieved better hold detection results on the new custom gym images. The image below shows the results of the preprocessing:

Figure 10: Image preprocessing results

3.3. Colour Extraction and Route Detection Results

In this section, the results for colour and route detection are provided for the algorithms described in the section above.

Figure 11: K-Means: colour extraction

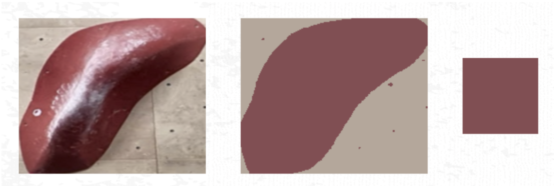


Figure 12: GMM: colour extraction

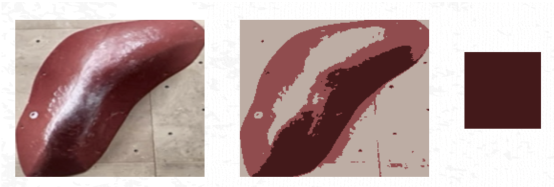


Figure 13: Mean-Shift: colour extraction

Figure 14: Hue HSV: colour extraction

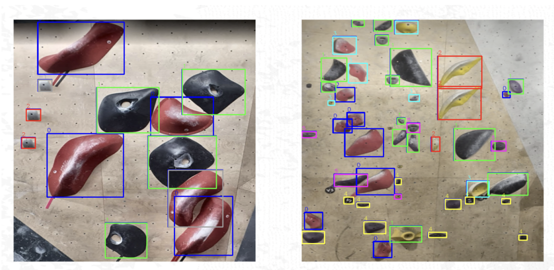


Figure 15: DBSCAN: route detection

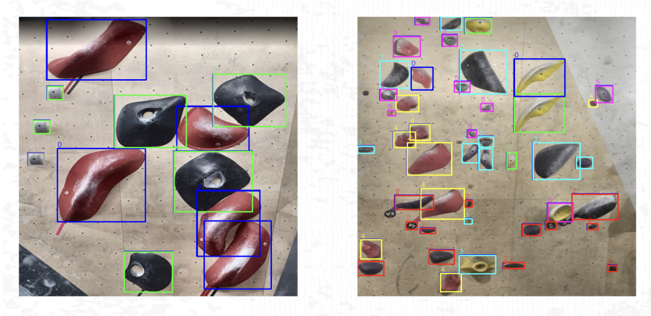


Figure 16: HDBSCAN: route detection

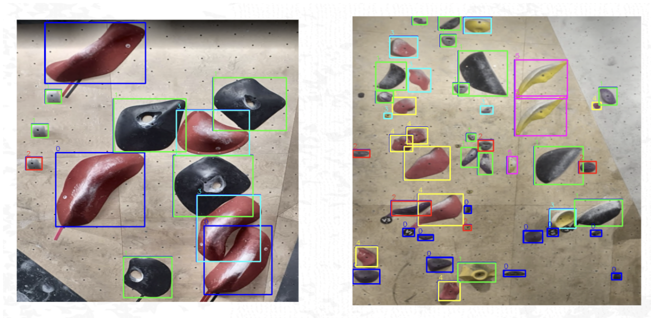


Figure 17: Agglomerative Clustering: route detection

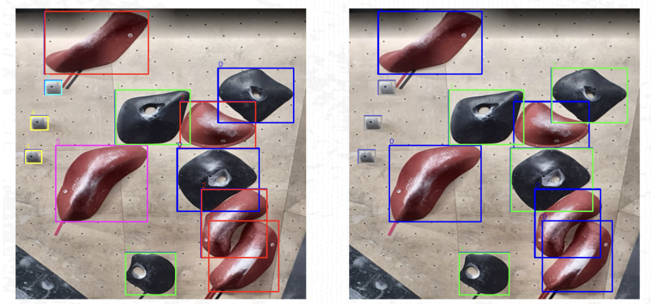


Figure 18: Agglomerative vs HDBSCAN: route detection

4. Discussion

The results of this project demonstrate that automated bouldering route detection is feasible when combining modern object detection models with classical clustering techniques. Across the given experiments (inference on the custom images), the YOLOv11 model demonstrated reliable results for hold detection. One of the aspects of the project was to focus on exploring the different YOLO model sizes, the feasibility of training them and performance. For example, from 3 it can be seen that, if trained correctly, the small YOLOv11 model can achieve significantly higher accuracy compared to the nano version. One of the factors is that the small version contains about 9 million parameters, whereas the nano one has about 2.5 million parameters. From that table, we can state that larger YOLO variants provided stronger accuracy at the cost of increased training and inference time. Also, as illustrated in Figures 4 and 5, the model performance consistently improved with additional training epochs. This can also be seen in the images: 7 — 9, where the larger models with more parameters or models trained for a larger number of epochs are able to achieve higher accuracy when detecting the holds on the given image.

Next, we can see that preprocessing has an important role in boosting detection quality. After applying techniques such as CLAHE contrast normalization and Gaussian blurring, we saw an improvement in the accuracy of hold detection, since they helped mitigate issues related to uneven lighting and chalk marks. Figure 6 shows how these preprocessing steps allowed YOLOv11 to not detect areas which were previously labelled as holds by the model. This reinforces the need for preprocessing when working with indoor climbing images, which frequently have poor lighting, shades and uneven hold colour.

After detecting the holds with the bounding boxes, the next step was to extract the colour of the hold. This task presented several challenges. Although K-Means, GMM, Mean-Shift, and HSV histogram methods were able to extract the dominant colours from the detected holds (images 11 — 14), the process was often noisy. Even after masking the background and applying preprocessing denoising algorithms, bounding boxes still included wall regions, chalk contamination, bolt holes, and shadows, which required the need to group semi-similar colours for better route detection. Furthermore, clustering methods struggled particularly with the holds sharing similar colours with the wall. This suggests the need for the use of the segmentation masks, which can significantly improve colour-based feature extraction.

For route detection, the clustering algorithms produced mixed results. DBSCAN, HDBSCAN, and Agglomerative Clustering each were able to group holds into spatially coherent clusters (Figures 15 — 17). We can see that for the simpler images, the HDBSCAN was able to detect the two full routes reliably. Notice that for the more realistic wall, none of the algorithms could consistently isolate full, correct routes. HDBSCAN performed best due to its ability to adapt to variable cluster densities and automatically separate noise points. DBSCAN was highly sensitive to the choice of **epsilon**, and Agglomerative Clustering tended to have too many clusters, as can be seen in the Figure 18. The results indicate that just these algorithms are insufficient for effective route detection. This is because two holds may be close in Euclidean space yet belong to different routes or, conversely, a valid sequence may contain holds that are spatially far apart but connected through feasible movement.

The challenge of route-level clustering points out an issue with the approach taken: bouldering routes are not directly vertical and contain a multitude of different movements. Climbers follow a path where each of their moves depends on body orientation, reachability, and intended movement flow. This indicates that purely unsupervised clustering algorithms, without the context of climbing movement, cannot capture the routes accurately. This is also indicative of the past research, where supervised sequential models (e.g., RNNs) outperform unsupervised clustering because they capture relational patterns between holds.

Overall, this project establishes a good foundation for automated climbing analysis. The pipeline successfully detects holds, extracts approximate colour information from the bounding boxes, and forms candidate route clusters. However, it also has certain limitations of the currently taken approaches: unsupervised clustering algorithms struggle to accurately detect the routes on a 'busy' wall and suggest the need for sequential and graph-based models. Further, the segmentation model could be a good alternative for getting accurate hold detection and assisting with the colour extraction. The present work serves as a necessary first step and motivates the development of the full intelligent route detection and generation system outlined in the Future Work section.

5. Conclusion

As climbing continues to grow in popularity and more gyms open worldwide, the demand for skilled route setters and creative bouldering problems design increases significantly. This highlights the growing need for automation in route detection, generation and analysis. The work presented in this project establishes a pipeline for detecting climbing holds, extracting their colours, and clustering them into potential bouldering routes. The implemented project combines a YOLO-based object detector with several colour-extraction algorithms (K-Means, Mean Shift, GMM, and HSV hue) and spatial clustering methods (DBSCAN, HDBSCAN, and Agglomerative Clustering).

The results demonstrate that the pipeline can successfully detect holds and group them by colour and spatial proximity, forming a foundation for automated route detection and analysis. The project also illustrates the limitations of purely unsupervised clustering approaches, particularly with overlapping routes and difficulties in clustering all holds spatially, reinforcing the need for sequential and graph-based models.

Overall, the system for automatically detecting routes, forms a basis for future work on automated climbing route generation. In the next stage, dataset augmentations, segmentation models, and graph-driven sequence learning will be integrated to move from route identification toward machine-generated climbs.

6. Future Work

From the results of this project, there are a few directions that can further enhance the performance, reliability, and applicability of an automated system for bouldering route detection and generation.

First, hold detection can be improved by transitioning from bounding-box–based detectors to instance segmentation models such as YOLO-Seg, Mask R-CNN, or SAM. Unlike bounding boxes, segmentation masks can provide precise hold boundaries, enabling more accurate colour extraction. This will help with colour classification since it will be performed directly on segmented masks rather than bounding-box pixels, reducing noise from overlapping annotations, chalk, and lighting.

In order to achieve higher accuracy with the YOLO object detection, in the future, we can apply custom data augmentations, through image preprocessing and augmentation: CLAHE contrast normalization and Gaussian blurring to reduce chalk patterns and different shading/lighting, which may increase robustness of hold detection across different climbing gyms.

Once the first step of the pipeline is robust, next major direction to explore will be route generation. After hold detection, the next step is to build a reachability graph that encodes spatial relations and human movement constraints. A Graph Neural Network (GNN) will be used to embed hold attributes. Next, a Transformer-based decoder can be trained to place holds sequentially, simulating realistic climbing movement patterns. This generative model will be evaluated with professional route setters to ensure feasibility and creativity of the produced routes.

Overall, these future improvements will help in creating a robust system that is capable of analyzing real climbing walls, extracting valid routes, and generating new climbs that respect human abilities.

Acknowledgements

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